Discriminant and Classification Analysis of Prostate Cancer Prognostic Measurements

STAT 419: Multivariate Analysis — Spring 2026 Course Project

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A Introduction / Background

A team of urologists set out to study how prostate-specific antigen (PSA) and a collection of clinical measurements relate to the progression of disease in men with advanced prostate cancer. The data set summarizes seven quantitative measurements taken on each of 97 such patients, along with a categorical variable indicating the severity of the disease. The goal of this project is to determine whether those seven measurements (or subset of them), considered jointly, can distinguish patients according to the severity of the seminal vesicle invasion, and to identify which measurements contribute most to that distinction.

The grouping variable for all analyses is **DIAGN**, which indicates the severity of the seminal vesicle invasion. This variable takes on three levels: low, moderate, and severe and it is nearly balanced with 32 low observations, 33 moderate observations, and 32 severe observations.

The remaining seven variables are all quantitative and will serve as the predictors:

- **PSA** - serum prostate-specific antigen level, in mg/ml. A widely used biomarker for prostate disease; elevated values are associated with more advanced disease.
- **Volume** - volume of the prostate cancer, in cubic centimeters (cm3). A larger tumor volume generally signals a more advanced condition.
- **Weight** - weight of the prostate gland, in grams (g).
- **Age** - age of the patient, in years.
- **BPH** - amount of benign prostatic hyperplasia, in square centimeters (cm2). A measure of non-cancerous enlargement of the prostate.
- **Capsule** - degree of capsular penetration, i.e. the extent to which the tumor has pushed into or through the prostatic capsule.
- **Gleason** - a pathologically determined qualitative grade of the disease. Higher scores indicate a worse prognosis; in this sample the score takes only the integer values 6, 7, 8.

The analysis using these features was in three stages. First we examined the distribution of each predictor through histograms and summary statistics (Section B). We then checked for multicollinearity amongst our features, and fitted linear discriminants to find the combination of variables that best separate the three severity groups while also testing these combinations for significance (Section C.2). Finally, using the most influential variables, we constructed linear classification functions and applied them to classify patients and evaluate how accurately the resulting rule determines the true severity groups (Section C.3).

B Graphs and Summary Statistics

Before analysis, we will see the nature of each variable in our dataset. Below is a summary statistics table for each variable.

```r
knitr::kable(data.frame(
  Variable = QUANT_VARS,
  Min      = sapply(cancer[QUANT_VARS], min, na.rm = TRUE),
  Q1       = sapply(cancer[QUANT_VARS], quantile, probs = 0.25, na.rm =
   ↪  TRUE),
  Mean     = sapply(cancer[QUANT_VARS], mean, na.rm = TRUE),
  Median   = sapply(cancer[QUANT_VARS], median, na.rm = TRUE),
  Q3       = sapply(cancer[QUANT_VARS], quantile, probs = 0.75, na.rm =
   ↪  TRUE),
  Max      = sapply(cancer[QUANT_VARS], max, na.rm = TRUE),
  SD       = sapply(cancer[QUANT_VARS], sd, na.rm = TRUE),
  row.names = NULL
), digits = 4)
```

Table 1

Variable	Min	Q1	Mean	Median	Q3	Max	SD
PSA	0.6510	5.6410	23.7301	13.3300	21.3280	265.0720	40.7829
Volume	0.2592	1.6653	6.9987	4.2631	8.4149	45.6042	7.8809
Weight	10.6970	29.3710	45.4914	37.3380	48.4240	450.3390	45.7051
Age	41.0000	60.0000	63.8660	65.0000	68.0000	79.0000	7.4451
BPH	0.0000	0.0000	2.5347	1.3499	4.7588	10.2779	3.0312
Capsule	0.0000	0.0000	2.2454	0.4493	3.2544	18.1741	3.7833
Gleason	6.0000	6.0000	6.8763	7.0000	7.0000	8.0000	0.7396

Below we can also see histograms of each variable.

```r
par(mfrow=c(3,3))

for (var in QUANT_VARS) {
  hist(cancer[[var]],
       main = paste("Histogram of", var),
       breaks = 15,
       xlab = var)
}
```

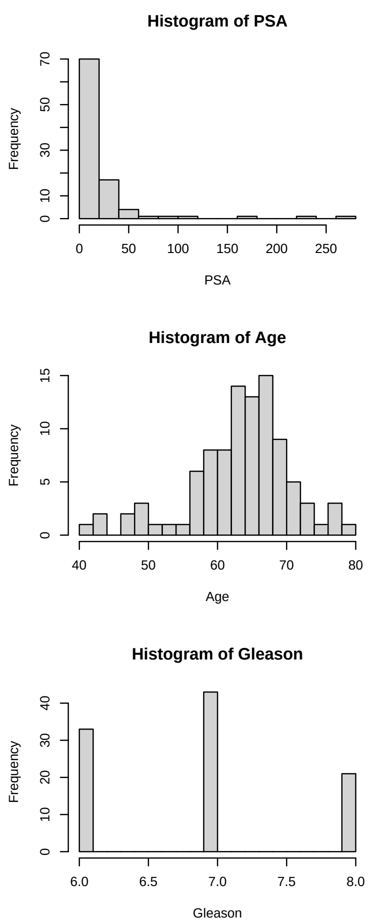

Figure 1: Histograms for all quantitative variables

Based on the histograms it looks like the variables `PSA`, `Volume`, `Weight`, `BPH`, `Capsule` are non normal. We can try to apply transformations on them to see if we can make them look more normal.

```r
par(mfrow=c(2,3))

# Variables requiring transformation (identified from raw histograms above)
TRANSFORMED_VARS <- c('PSA', 'Volume', 'Weight', 'BPH', 'Capsule')
LOG_VARS <- c('PSA', 'Volume', 'Weight')          # strictly positive
LOG1P_VARS <- setdiff(TRANSFORMED_VARS, LOG_VARS) # contain zeros

for (var in LOG_VARS) {
  hist(log(cancer[var][,]), main=paste("Histogram of LOG(", var, ")",
   ↪  sep=''), breaks=15, xlab=paste('log(',var,')',sep=''))
}

for (var in LOG1P_VARS) {
  hist(log1p(cancer[var][,]), main=paste("Histogram of LOG(", var, " + 1)",
   ↪  sep=''), breaks=15, xlab=paste('log(',var,' + 1)',sep=''))
}

plot.new()  # pad the 2x3 grid (5 plots, 1 empty cell)
```

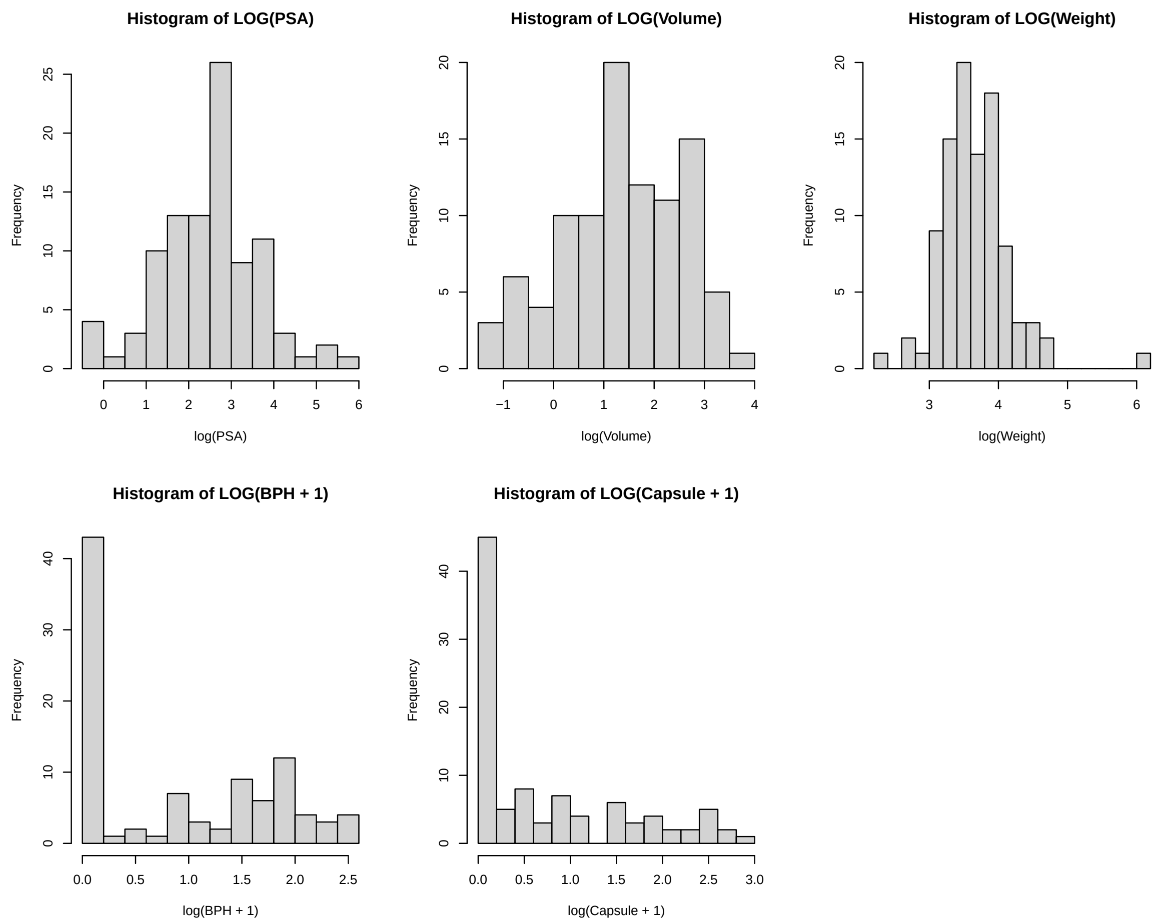

Figure 2: Histograms for all transformed variables

> **Note**
>
> `BPH` and `Capsule` are transformed with a log + 1 transformation because they contain 0 values. See Table 1

Looking at Figure 2, `PSA`, `Volume`, `Weight` all appear to be roughly normal, however `BPH` and `Capsule` do not. Because discriminant analysis assumes approximate multivariate normality we drop them from our data and only use `log(PSA)`, `log(Volume)`, `log(Weight)`, and the untransformed `Age` and `Gleason` for our analysis.

Below is a scatterplot matrix of the transformed dataset.

```r
# Build the modelling dataset
cancer.mod <- cbind(cancer[setdiff(colnames(cancer), TRANSFORMED_VARS)],
 ↪  log(cancer[LOG_VARS]))

# Color points by DIAGN group; exclude DIAGN column from the numeric panels
pairs(cancer.mod[, -1],
      main = "Scatterplot Matrix of Quantitative Variables",
      pch = 19, cex = 0.5,
      col = c("steelblue", "darkorange", "firebrick")[cancer.mod$DIAGN])
```

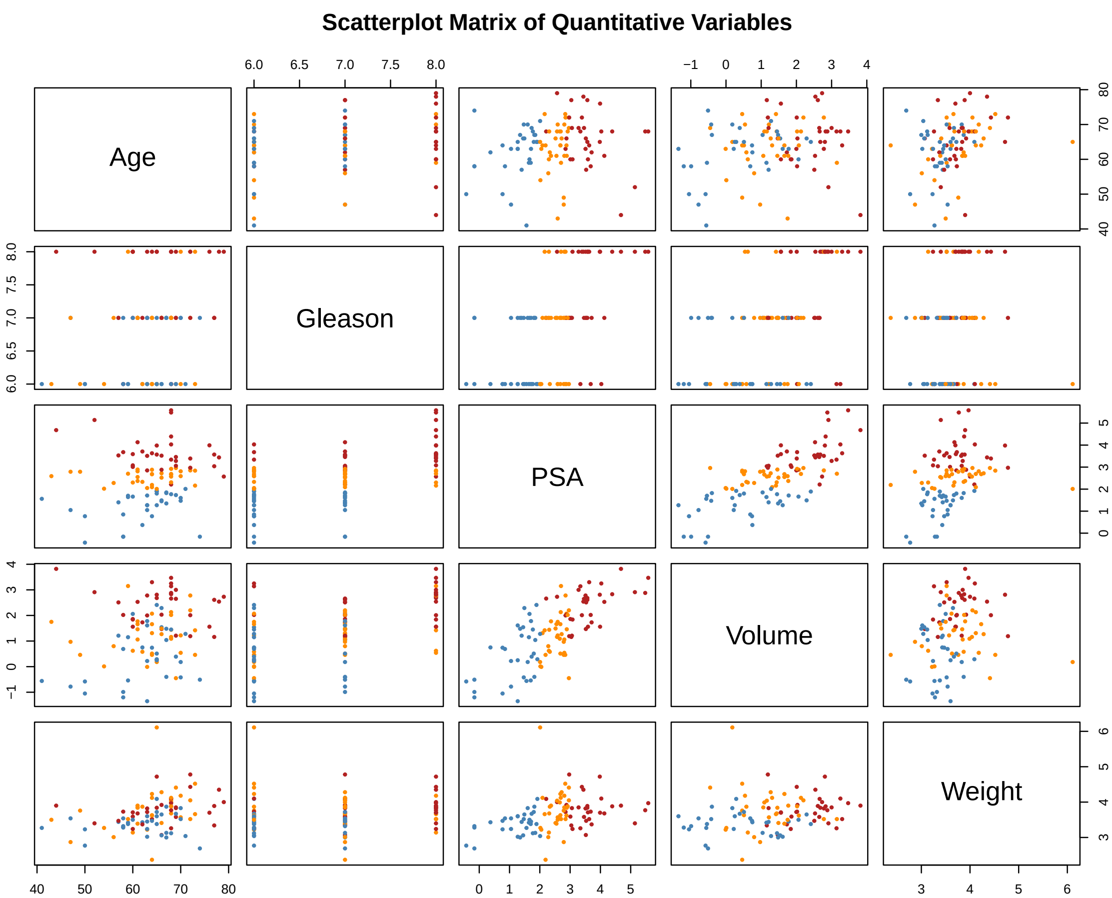

Figure 3: Scatterplot matrix of transformed dataset

B.1 Outlier Identification

After looking at the data we can see some trends and possibly some outliers. Of note there is a point that appears to have a higher weight than the others. We can also check for outliers by performing a slightly more statistical method. AssAssuming approximate multivariate normality, we can see how far away each dot is from the center, and the probability of seeing an observation that extreme.

```
# Use mahalanobis to measure distance from centroid
md <- mahalanobis(
  x      = cancer.mod[, -1],
  center = colMeans(cancer.mod[, -1], na.rm = TRUE),
  cov    = cov(cancer.mod[, -1], use = "complete.obs")
)

# Bonferroni correction
sig_level <- 0.05 / nrow(cancer[, -1])

# Upper-tail p-value: probability of a chi-sq value this large under MVN
# (large distance -> small p -> flagged as outlier)
observation_p <- pchisq(md, df = ncol(cancer.mod[, -1]), lower.tail = FALSE)

knitr::kable(cbind(cancer, observation_p)[which(observation_p < sig_level),
 ↪  ])
```

Table 2

	DIAGN	PSA	Volume	Weight	Age	BPH	Capsule	Gleason	observation_p
41	Moderate	7.463	1.1972	450.339	65	5.4739	0	6	6.9e-06

Based on the results we can see that observation 41 appears to be an outlier. Looking at the output of Table 2 and comparing it to Table 1, all of the variables seem to be within typical range observed in this data set, with the exception of `Weight`. The weight of 450.339 grams is significantly higher than the normal range. In fact, it is 8.858 standard deviations away from the mean weight. For this reason we excluded this data point from our analysis.

```
cancer.mod <- cancer.mod[-41,]
```

We then checked our MVN assumptions (see Section B.2).

B.2 MVN Assumptions

```
par(mfrow=c(2,2))

check.mvnorm.plot(cancer.mod[,-1])
mtext("All Values",side=3)
check.mvnorm.plot(cancer.mod[cancer.mod$DIAGN == "Low",][,-1])
mtext("DIAGN=Low",side=3)
check.mvnorm.plot(cancer.mod[cancer.mod$DIAGN == "Moderate",][,-1])
mtext("DIAGN=Moderate",side=3)
check.mvnorm.plot(cancer.mod[cancer.mod$DIAGN == "Severe",][,-1])
mtext("DIAGN=Severe", side=3)
```

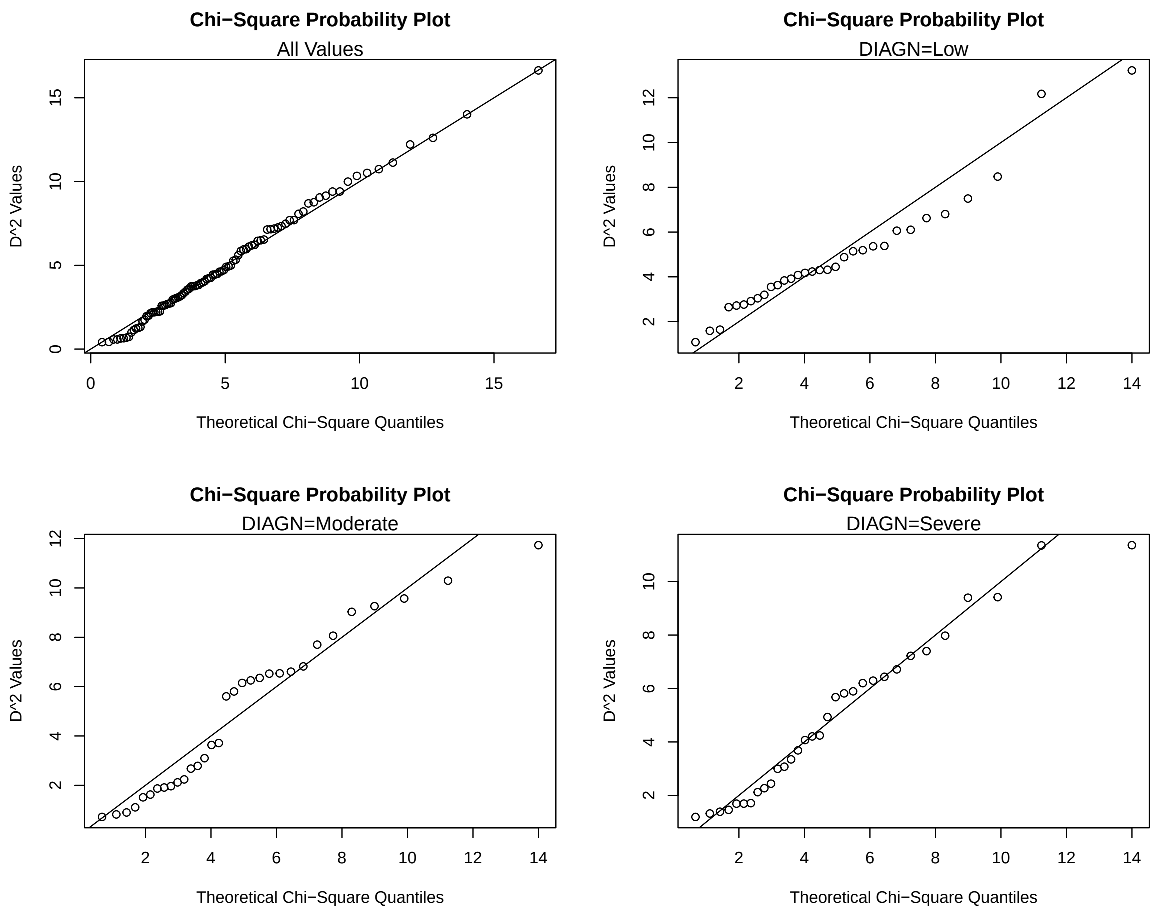

Figure 4: MVN Q-Q Plots

Table 3

```r
# Box's M tests H0: all groups share a common covariance matrix (required for
↪  LDA)
cov_mats <- list(
  Low      = cov(cancer.mod[cancer.mod$DIAGN == "Low",      -1], use =
   ↪  "complete.obs"),
  Moderate = cov(cancer.mod[cancer.mod$DIAGN == "Moderate", -1], use =
   ↪  "complete.obs"),
  Severe   = cov(cancer.mod[cancer.mod$DIAGN == "Severe",   -1], use =
   ↪  "complete.obs")
)

n_obs <- unname(as.vector(table(cancer.mod$DIAGN)[c("Low", "Moderate",
 ↪  "Severe")]))

Box.M(cov_mats, n_obs)
```

```
$u
[1] 73.14333

$p.value.u
[1] 1.819254e-05

$chi.df
[1] 30
```

These plots look pretty much like they're all MVN. The last check is to make sure they all have the same covariance matrices. To do this we use Box's M test.

> **Caution**
>
> We actually failed Box's M test. However, we still decided to proceed with discriminant analysis as it looks close enough.

Based on the output it appears all groups follow a multivariate normal distribution. This allows us to move on to discriminant analysis.

C Discriminant and Classification Analysis

C.1 Correlated Quantitative Variables (Multicollinearity)

```r
knitr::kable(cor(cancer.mod[,-1]))
```

Table 4

	Age	Gleason	PSA	Volume	Weight
Age	1.0000000	0.2294805	0.1707274	0.2278361	0.3470233
Gleason	0.2294805	1.0000000	0.5384069	0.5068264	0.2226661
PSA	0.1707274	0.5384069	1.0000000	0.7340158	0.4353842
Volume	0.2278361	0.5068264	0.7340158	1.0000000	0.2857982
Weight	0.3470233	0.2226661	0.4353842	0.2857982	1.0000000

To assess for multicollinearity, we analyze the correlation coefficients for each pair of quantitative variables. Our scatterplot matrix shows that there is no strong linear relationship between any two variables. We can then look at the calculated correlation coefficient table. Using a correlation threshold of $|r| \geq 0.80$, none of our variables have high correlation. We will not flag any of our variables for multicollinearity. We will keep all of our 5 quantitative variables.

C.2 Discriminant Analysis

Discriminant Functions and Variable Importance

```r
da <- discrim(cancer.mod[,-1], cancer.mod[,1])

n_ld     <- ncol(da$a.stand)
std_coef <- da$a.stand
rownames(std_coef) <- colnames(cancer.mod[,-1])
colnames(std_coef) <- paste0("LD", seq_len(n_ld))  # label columns LD1, LD2,
 ↪  ...

knitr::kable(std_coef)
```

Table 5

	LD1	LD2
Age	-0.0478356	0.1523041
Gleason	-0.1538214	0.1146393
PSA	-0.5899614	-0.2450616
Volume	-0.0681763	0.4576420
Weight	-0.0426631	-0.2893474

A linear discriminant analysis was performed to observe possible separation between the groups of severity of seminal vesicle invasion and which variables contribute the most in that separation. The discriminant analysis resulted in 2 discriminant functions:

$$LD1 = -0.0478\,Age - 0.1538\,Gleason - 0.5900\,\log(PSA) - 0.0682\,\log(Volume) - 0.0427\,\log(Weight)$$

$$LD2 = 0.1523\,Age + 0.1146\,Gleason - 0.2451\,\log(PSA) + 0.4576\,\log(Volume) - 0.2893\,\log(Weight)$$

When looking at the importance of each variable in the first discriminant function, we get:

```r
# Sort variables by absolute LD1 coefficient (largest = most important)
ld1_order <- names(sort(abs(std_coef[, 1]), decreasing = TRUE))

knitr::kable(
  data.frame(
    Variable = ld1_order,
    LD1 = std_coef[ld1_order, 1],
    row.names = NULL
  ),
  digits = 4,
)
```

Table 6

Variable	LD1
PSA	-0.5900
Gleason	-0.1538
Volume	-0.0682
Age	-0.0478
Weight	-0.0427

The table of the standardized coefficients allows us to see that PSA seems to strongly contribute to group separation of severity of seminal vesicle invasion. Then Gleason seems to have moderate contributions while Volume, Age, Weight seem to have the weakest contribution.

Test of Significance of the Discriminant Functions

To see which linear discriminant functions are significant in group separation, we will test:

(1) $H_0 : \alpha_1 = \alpha_2 = 0$, H_a : at least one $\alpha \neq 0$

(2) $H_0 : \alpha_2 = 0$, $H_a : \alpha_2 \neq 0$

Since we have two discriminant functions, 2 tests were performed (listed above), we must adjust our level of significance to $\frac{0.05}{2} = 0.025$. The results are:

```
discr_sig_tbl <- discr.sig(cancer.mod[,-1], cancer.mod[,1])
knitr::kable(discr_sig_tbl)
```

Table 7

	Lambda	V	p.values
LD1	0.2456491	127.750456	0.0000000
LD2	0.9476667	4.891468	0.2986159

LD1: With a p-value of 0.0000 being less than our adjusted significance level of 0.025, we have enough evidence that the first linear discriminant function works well in group separation of severity of seminal vesicle invasion.

LD2: With a p-value of 0.2986 being greater than our adjusted significance level of 0.025, we don't have enough evidence that the second discriminant function is useful in group separation of severity of seminal vesicle invasion.

Significance of Each Variable, Adjusted for the Others

After seeing the standardized coefficients of the first discriminant function, the variable PSA might be important in group separation. Here are the formal partial F tests for each variable:

H_0: Variable (y) does not contribute to group separation of severity of seminal vesicle invasion, adjusting for the other variables.

H_a: Variable (y) does contribute to group separation of severity of seminal vesicle invasion, adjusting for the other variables.

Since there are 5 tests being performed, we must adjust our significance level to $\alpha^* = \frac{.05}{5} = 0.01$.

Table 8

```r
partial_f_tbl   <- partial.F(cancer.mod[,-1], cancer.mod[,1])
n_partial_tests <- nrow(partial_f_tbl)  # one test per variable
```

```r
knitr::kable(partial_f_tbl)
```

Table 9

	Lambda	F.stat	p.value
PSA	0.5688174	33.7324827	0.0000000
Gleason	0.9625959	1.7291599	0.1833405
Volume	0.9672944	1.5046102	0.2276985
Weight	0.9855378	0.6530123	0.5229511
Age	0.9934851	0.2918139	0.7476194

From our results, PSA significantly contributes the most in group separation of severity of seminal vesicle invasion, adjusting for the other variables.

All other variables do not individually contribute significantly in group separation of severity of seminal vesicle invasion, after adjusting for the other variables.

Plot of the First Two Discriminant Functions

Below is a plot of the two linear discriminant functions.

```r
discr.plot(cancer.mod[,-1], cancer.mod[,1], title='Discriminant Plot for
 ↪  Diagnosis Groups', leg = levels(cancer.mod[,1]))
```

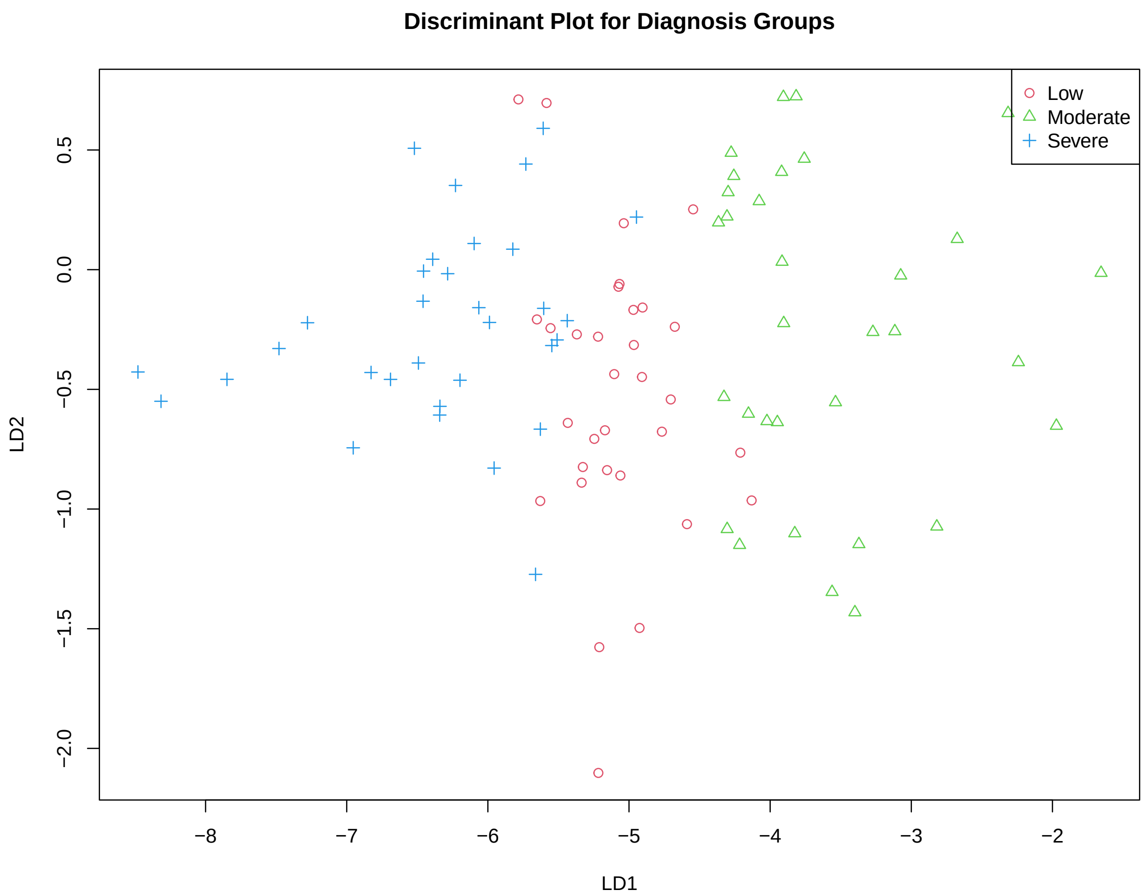

Figure 5: Discriminant Plots

From this plot, it is visually evident the first discriminant function (LD1) captures the group separation well of the three groups of low, moderate, and severe seminal vesicle invasion. The severe group of seminal vesicle invasion is on the left while the moderate level is on the right. Low severity of seminal vesicle invasion seems grouped in between the moderate and severe levels.

The second discriminant function (LD2) does not capture the separation among the groups as projections of the points on the line are clumped together. This further supports that LD1 is the better discriminant function that works well in group separation.

C.3 Classification Analysis

For classification analysis we first select the top 4 most important variables from the LD1 importance ranking.

Looking at Table 6, the top four variables, in order of importance, are PSA, Gleason, Volume, Age.

Linear Classification Functions

The linear classification function for each severity group is given below. An observation is assigned to the group whose function returns the largest score.

```r
lc <- lin.class(cancer.mod[, top_vars], cancer.mod[, 1])

# Reshape coefficients: rows = variables, columns = groups
coefs_byvar <- t(lc$coefs)
rownames(coefs_byvar) <- top_vars
colnames(coefs_byvar) <- levels(cancer.mod[,1])

# Prepend the constant term (c0) as its own row
lcf_table <- rbind(Constant = lc$c.0, coefs_byvar)
colnames(lcf_table) <- levels(cancer.mod[,1])

knitr::kable(lcf_table, digits = 4)
```

Table 10

	Low	Moderate	Severe
Constant	-82.7467	-94.7592	-111.0073
PSA	3.1830	6.7816	9.0045
Gleason	15.6641	16.2782	17.1394
Volume	-3.7045	-3.9340	-3.2372
Age	1.0121	1.0323	1.0624

Written out explicitly, where 1 = Low, 2 = Moderate, and 3 = Severe:

$$L_1(y) = 3.1830\,\log(PSA) + 15.6641\,Gleason - 3.7045\,\log(Volume) + 1.0121\,Age - 82.7467$$
$$L_2(y) = 6.7816\,\log(PSA) + 16.2782\,Gleason - 3.9340\,\log(Volume) + 1.0323\,Age - 94.7592$$
$$L_3(y) = 9.0045\,\log(PSA) + 17.1394\,Gleason - 3.2372\,\log(Volume) + 1.0624\,Age - 111.0073$$

Classifying Observation #1

We apply the three functions to the first observation, compute each score, and assign the group with the largest value.

```
groups <- levels(cancer.mod[,1])

# Compute LCF score for a single observation against each group
lcf_scores <- function(x_vec) {
  sapply(groups, function(g) lc$c.0[match(g, groups)] + sum(coefs_byvar[, g]
   ↪  * x_vec))
}

obs1_x      <- cancer.mod[1, top_vars]         # predictor values for
 ↪  observation 1
obs1_scores <- lcf_scores(obs1_x)              # score against each group's
 ↪  function
obs1_pred   <- groups[which.max(obs1_scores)]  # predicted group = highest
 ↪  score
obs1_actual <- as.character(cancer.mod[1, 1])  # true group label

knitr::kable(
  data.frame(Group = groups, Score = round(obs1_scores, 4), row.names =
   ↪  NULL),
  digits = 4,
)
```

Table 11

Group	Score
Low	108.3140
Moderate	110.5630
Severe	108.5532

The largest score belongs to the **Moderate** group, so Observation #1 is classified as Moderate. Its actual group is Moderate, so this prediction is correct.

Confusion Matrix and Error Rates

Using resubstitution (apparent) classification on all observations:

```r
cls       <- rates(cancer.mod[, top_vars], cancer.mod[,1], method = "l")
confusion <- cls[["Confusion Matrix"]]
ACCR      <- cls[["Correct Class Rate"]]
APER      <- cls[["Error Rate"]]

knitr::kable(as.data.frame.matrix(confusion))
```

Table 12

	Low	Moderate	Severe
Low	29	3	0
Moderate	0	29	3
Severe	0	7	25

```r
n         <- nrow(cancer.mod)
n_correct <- round(ACCR * n)

knitr::kable(
  data.frame(
    Metric = c("Apparent Correct Classification Rate (ACCR)",
               "Apparent Error Rate (APER)", "N", "Correct"),
    Value  = c(ACCR, APER, n, n_correct)
  ),
  digits = 4
)
```

Table 13

Metric	Value
Apparent Correct Classification Rate (ACCR)	0.8646
Apparent Error Rate (APER)	0.1354
N	96.0000
Correct	83.0000

We have an overall correct classification rate of 86.5%, indicating a good discrimination model. The 13.5% of misclassifications fall between adjacent severity groups; there are no misclassifications between the Low and Severe groups. With a high correct classification rate, the model provides accurate prediction of the severity groups.

D Summary

The goal of this project was to determine whether 7 clinical measurements could effectively distinguish prostate cancer patients according to the severity of seminal vesicle invasion. Based on our analysis, the answer is yes. Among the measured variables, PSA was the most influential in separating the groups, as evidenced by its significance in the discriminant analysis (Section C.2) and its large contribution to the first linear discriminant function. In addition, plotting the observations with respect to LD1 and LD2 (Figure 5) revealed strong separation among the severity groups, particularly along LD1.

This conclusion is further supported by the classification results (Section C.3). Using the linear classification functions, the apparent correct classification rate (ACCR) was 0.8646, meaning that 83 of the 96 observations were correctly classified. The confusion matrix (Table 12) also indicates that the model avoided the most severe types of misclassification: no patients in the low-severity group were classified as severe, and no patients in the severe group were classified as low. Instead, the misclassifications were limited to adjacent severity categories, such as low-severity observations being classified as moderate. Overall, these results suggest that the clinical measurements, particularly PSA, provide meaningful discrimination among levels of seminal vesicle invasion severity.

One notable result was that no variables were collinear enough to drop in Section C.1, even though we expected PSA, Volume, and Capsule might have been redundant; their correlations were moderate, but never reached our thresholds.

If we were to repeat this project, we would make two key improvements. First, we would explore alternative data transformations, either in place of or in combination with log transformations, such as the Box–Cox transformation. This would be particularly useful for variables such as BPH and Capsule, which remain skewed even after applying a log transformation. Second, we would implement leave-one-out cross-validation to estimate the classification error rate. This would allow us to assess whether the selected transformations and resulting classification model generalize well to new observations, rather than overfitting to the particular dataset used in this analysis.

Finally, one variable we wish the data would have included is a direct measure of tumor stage (such as a TNM clinical stage) or a PSA reading taken over time rather than at a single point. More generally, variables that would give information about the trajectory of the disease for a given patient would help acheive cleaner separation for the observations on the border of moderate and severe, which were the toughest to classify with measurements available.